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LARGE-SCALE HORIZONTAL DISPLACEMENTS IN SOUTHERN AUSTRALIA — CONTRARY EVIDENCE FROM PALAEOMAGNETISM

by J. W. GIDDINGS and B. J. J. EMBLETON

(With 1 Table and 3 Figures)

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ABSTRACT

Palaeomagnetic results from 40 dykes sampled on Eyre Peninsula and Yorke Peninsula, South Australia, are inconsistent with the clockwise rotation of the 'foot' of Yorke Peninsula by fault-drag during the Early Ordovician. A reinterpretation of earlier data from Tasmania also indicates that the east-west structural trends in the Mount Reed volcanic arc and Dundas Trough are primary. The effects of shearing were probably restricted to linear displacements without causing rotation of adjacent structural elements. Late Precambrian and Cambrian palaeomagnetic data from Fleurieu Peninsula and Kangaroo Island have also been analysed. They do not clarify the pattern of structural evolution in the southern part of the Adelaide Orogen because the magnetic remanence was probably acquired syntectonically.

INTRODUCTION

Crawford & Campbell (1973*a*) postulated large-scale intra-continental shearing within Australo-Antarctica culminating in the Ordovician. One major shear forms what is now the southern boundary of Australia, and of the Gawler Block in part—see Figure 1. They regarded the southern part of the Adelaide Orogen as having been bent in a westerly direction by large-scale fault-drag along this shear. This appeared to provide the first satisfactory explanation of that deflection. They went on to suggest that the effects of gradual bending, which ultimately produced east-west structural trends in Kangaroo Island and the Fleurieu Peninsula, also involved the south-eastern part of the Gawler Block, 'slightly rotating the leg of Yorke Peninsula and, as the southernmost part of the Orogen was bent through ninety degrees, snapped the extremity of the Block round into an east-west position . . .', that is, producing the 'foot' of Yorke Peninsula. The arcuate shape of the Dundas Trough—Mount Reed volcanic arc in Tasmania was also attributed to deformation caused by shearing. They postulated that those features were originally linear, and bent by subsequent compression associated with the shear. Opposing views (Daily *et al.*, 1973; Harrington *et al.*, 1973) also based on geological arguments, maintain that the structural features are primary.

A combined program of palaeomagnetic and geochronological studies of Precambrian dyke swarms in Western and South Australia is near completion. It was designed primarily to investigate relationships between the major Precambrian blocks of southern Western Australia and South Australia for the times of the emplacement of the dykes. During the main study it has been possible to check palaeomagnetically the suggestion of rotation of parts of Yorke Peninsula relative to Eyre Peninsula. Rock samples were collected from 13 dykes which outcrop on the north and west coasts of the 'foot' of Yorke Peninsula and from 6 dykes on the west coast of the 'leg' of the Peninsula—see Figure 2. Twenty-one dykes were sampled on the east coast of southern Eyre Peninsula. All the dykes intrude crystalline basement and are Precambrian in age. Since they antedate the proposed Lower Palaeozoic shear, differences in magnetic declinations between each of the three sampling localities will provide estimates of the amounts of rotation if it ocurred at all.

Many palaeomagnetic directions have also been determined from late Precambrian and Lower Palaeozoic rock formations from widely separated localities throughout Australia (McElhinny & Luck, 1970; Luck, 1972; Embleton, 1972*a*, *b*; Embleton & Giddings, 1974). They support the contention that the main Australian platform has retained its gross structural unity since at least late Precambrian time (Embleton & Giddings, 1974). Earlier studies of late Precambrian and Cambrian sedimentary rock formations collected from Kangaroo Island, the southern Mount Lofty Ranges, and the Dundas Trough

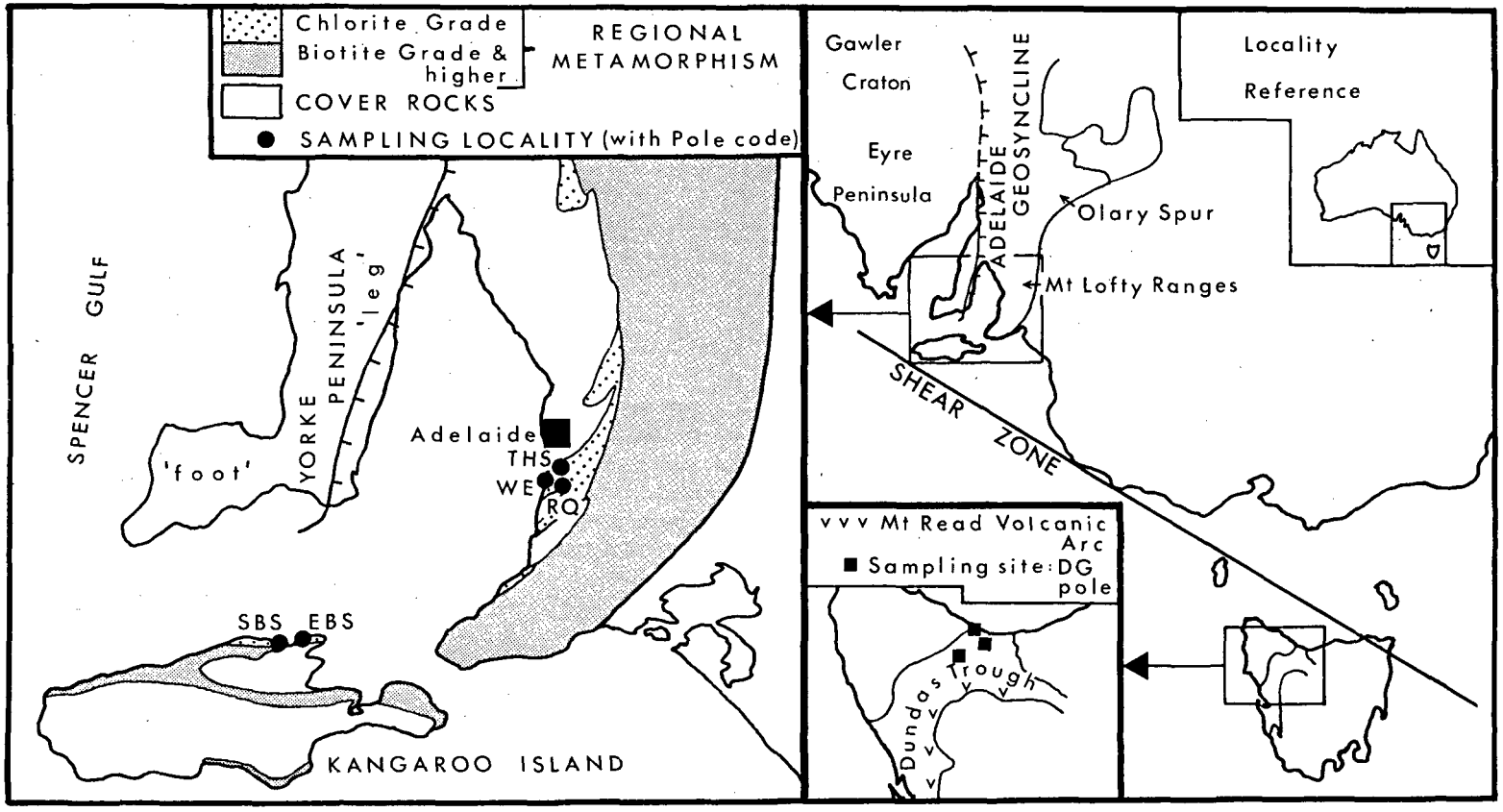


Fig. 1. The principal structural elements in South Australia and Tasmania which according to the Crawford & Campbell (1973*a*) hypothesis initially occupied meridional trends. Sampling localities for the earlier palaeomagnetic investigations are indicated. THS—Tapley Hill Slate, RQ—Marinoan Sandstone (Reynella Quarries), WE—Marinoan Sandstone (Wilmington equivalents), SBS—Stokes Bay Sandstone and Smith Bay Shale, EBS—Emu Bay Sandstone, DG—Dundas Group.

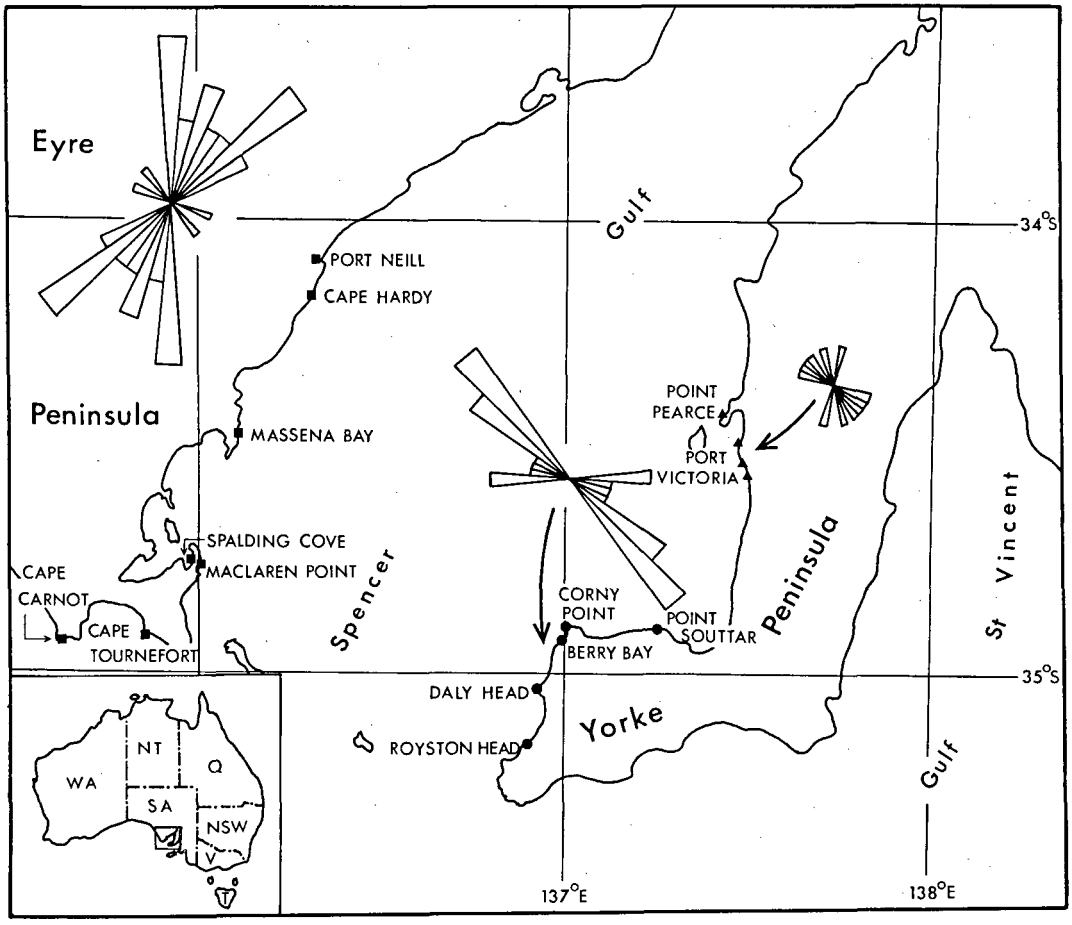


Fig. 2. Precambrian dyke sampling localities in the southeastern region of the Gawler Block. The inset rose diagrams depict measured dyke trends in the principal sampling regions. The shortest unit represents the trend of a single dyke and the longest unit represents four dykes. Each sector covers 10°.

Briden, 1964, 1967*a, b*) were considered inconclusive at the time of publication. A framework therefore now exists within which those data may be reviewed.

RESULTS FROM THE PRECAMBRIAN DYKES

The remanent magnetism acquired by the dykes has been tested for stability with the alternating field demagnetization technique (As & Zijderveld, 1958). Of the 40 dykes sampled, 34 have retained a proportion of their primary component of magnetic remanence. A full appreciation of the palaeomagnetic results and their interpretation in terms of Precambrian apparent polar wander is being prepared separately for publication. Table I contains the overall mean directions of remanent magnetization of the principal populations of dyke-means from each locality. Dyke-mean directions are plotted stereographically (on a Wulff net) in Figure 3.

(a) *Eyre Peninsula.* Directions of remanent magnetization plot in two distinct groups. The group-means are not antiparallel and probably record successive directions of the geomagnetic field acquired during two separate periods of dyke-intrusion.

(b) *Yorke Peninsula* (the 'foot'). One principal group of directions is apparent; one dyke gave an approximately antiparallel direction. Whether it reflects a geomagnetic field reversal or should strictly be considered a member

TABLE I

Summary of the palaeomagnetic data from Yorke Peninsula and Eyre Peninsula

Sampling Locality	N*	R	D°	I°	A_{95}
Eyre Peninsula					
(a) positive group	9	8.389	270	+59	15°
(b) negative group	10	9.665	032	−48	9°
Yorke Peninsula					
(a) 'leg'	5	4.768	275	+60	19°
(b) 'foot'	8	7.379	275	+65	17°

* N = number of dykes used to compute the mean, R = resultant of N unit vectors, D° = declination, I° = inclination and A_{95} = semi-angle of the cone of confidence at the 95% probability level (Fisher, 1953).

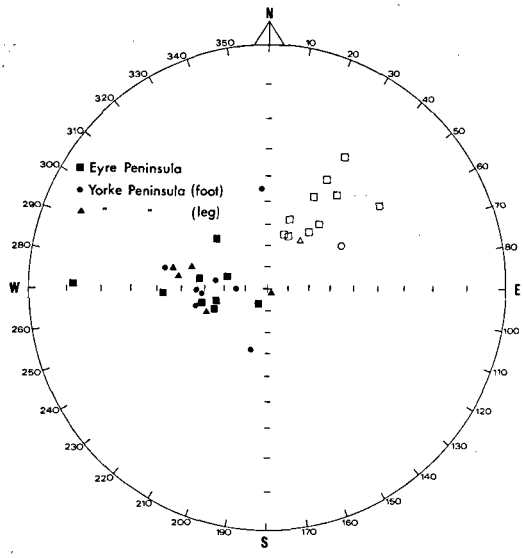


Fig. 3. Dyke-mean directions of primary magnetic remanence. The squares, circles, and triangles correlate with the sampling localities shown in Fig. 2. The positive directions (down-pointing, full symbols) obtained from the 'foot' and the 'leg' of Yorke Peninsula and Eyre Peninsula form a tight homogeneous group.

of the negatively magnetized group defined from Eyre Peninsula is uncertain.

(c) *Yorke Peninsula* (the 'leg'). Six dykes, 5 of which are positively magnetized and 1 negatively magnetized, form a group coincident with dyke-mean directions from Eyre Peninsula and the 'foot' of Yorke Peninsula.

Crawford & Campbell (1973*a*) report that lineations thought to be bedding foliation in the basement gneisses of the 'foot' of Yorke Peninsula diverge by up to 90° from lineations measured in the 'leg' and in the basement gneisses of Eyre Peninsula—following work by Crawford (1965).

Where the outcrops permitted, the structural trends of the dykes sampled for palaeomagnetism were measured. The compass rose diagrams, inset in Figure 2, show quite clearly that the dykes which intrude the basement gneisses of Yorke Peninsula conform to a unimodal distribution. Those intruding the 'foot' vary in strike from about 270° to 320° and intruding the 'leg' vary from 290° to 340°, with one dyke trending approximately north-south. On the other hand dyke trends on Yorke Peninsula are different from those on Eyre Peninsula. As seen in Figure 2, dykes on Eyre Peninsula generally trend between 0° and about 50°.

APPRAISAL OF PREVIOUS INVESTIGATIONS OF LATE PRE-CAMBRIAN-CAMBRIAN SEDIMENTS

Palaeomagnetic results have been described by Briden (1964) from the Upper Glacial and Interglacial beds of the late Precambrian Umberatana Group sampled south of Adelaide (Fig. 1), and from red-beds of Early and Middle Cambrian age sampled on Kangaroo Island. Much of the material studied was considered '. . . remagnetized, not in recent times, but as long ago as the Early Tertiary' (Briden, 1964, p. 98). This conclusion was reached for two principal reasons: (i) the uncleaned directions of magnetization were considered steeper than the present field direction and clustered around the Early Tertiary field direction—as it was then defined (Irving, 1964); and (ii) the postulated reheating effect of basalts, at that time thought to be of Early Tertiary age, which outcrop on low hills above the Cambrian outcrops on Kangaroo Island. To influence the thermal regime on Fleurieu Peninsula, the basalts would have to have been much more extensive. We now know from potassium-argon geochronology (Wellman, 1971) that the Kangaroo Island basalts are significantly older—about 170 m.y.—than was previously thought. An Early Tertiary remagnetization of the late Precambrian and Cambrian sediments therefore loses credence. Briden reported the results in meticulous detail because he acknowledged that '. . . some of these results will become more intelligible when further work is carried out on rocks of these ages in Australia' (Briden, 1964, p. 99).

The analysis was undertaken initially because our current research program involved the investigation of unmetamorphosed Cambrian

and late Precambrian sediments from the northern Flinders Ranges (Embleton & Giddings, 1974). Our study of the palaeomagnetism of the Pound Quartzite indicated that it was necessary to follow very closely the response of individual specimens to thermal cleaning. Since specimens cut from a sample often covered the stability spectrum, analysis of the data obtained at the sample level produced a confused pattern. Luck (1972) also encountered similar problems during his investigations of Lower Palaeozoic formations from northern Australia. Aware that remagnetization and the acquisition of viscous remanent magnetization may take place inhomogeneously we scrutinized the earlier results and tentatively identified specimens which after partial thermal demagnetization appeared to contain a stable component of magnetization oblique to the present field axis. The formations which yielded the data are the Tapley Hill Slates (older than the Marinoan glacials), Marinoan Sandstones from the Fleurieu Peninsula (as old as or older than the glacials) and the Emu Bay Sandstone, the Stokes Bay Sandstone, and the Smith Bay Shale of Early-Middle Cambrian age from Kangaroo Island. However, the directions of magnetic remanence obtained by Briden (1964), although scattered, were not unlike those obtained from Cambro-Ordovician formations described by Embleton (1972*b*), Luck (1972), and Embleton & Giddings (1974). It appears that the formations in the southern part of the Adelaide Orogen may well have suffered remagnetization, but it was a Cambro-Ordovician rather than a Tertiary event. Regional metamorphism and intrusion of granitic bodies in the southern Mount Lofty Ranges and Kangaroo Island occurred during the Delamerian Orogeny in the Late Cambrian and Early Ordovician (Offler & Fleming, 1968; Thomson, 1969). The localities Briden sampled for his investigation are situated within the chlorite zone of the greenschist facies. We believe that the regional metamorphism created an environment suitable for remagnetization.

IMPLICATIONS FOR LARGE-SCALE HORIZONTAL DISPLACEMENTS IN SOUTHERN AUSTRALIA

We interpret the directions of magnetization measured in the dykes as records of the ambient geomagnetic field direction at the time of dyke intrusion. Alternatively those dykes which constitute the positive (down-pointing) group of directions (Fig. 3) could be regarded as having been selectively remagnetized after the postulated rotation took place, i.e. after Early Ordovician. This interpretation is considered untenable: (a) the direction of magnetization is unlike any field direction yet measured from post-Ordovician rocks in Australia, (b) the bimodal distribution of directions has been retained, which would be extremely unlikely after remagnetization. Therefore, results from the southeastern region of the Gawler Block are inconsistent with a structural model involving local rotation in Yorke Peninsula (viz. the 'foot' with respect to the 'leg') and rotation of the entire Peninsula relative to Eyre Peninsula. Indeed, the consistency of the palaeomagnetic data may be considered particularly good and indicates that the region has maintained its physical integrity at least since the time of dyke intrusion. Furthermore, the trends of the dykes in the 'foot' and the 'leg' of Yorke Peninsula lend no support to the rotation hypothesis.

The result, acknowledged by Crawford & Campbell (1973*b*), only indicates that the southeastern margin of the Gawler Craton has remained structurally intact. The data cannot be extrapolated to disprove bending of the southern Mount Lofty Ranges since they constitute a distinct tectonic unit. Unfortunately, Briden's (1964) data are inconclusive in this context. If the beds were remagnetized, as we believe, then the magnetic remanence would have been acquired syntectonically. Unknown variables in attempting to elucidate this problem include the time relationships of folding and metamorphism (Offler & Fleming, 1968), the degree of remagnetization, and whether secondary components of magnetization were completely removed by cleaning the samples in the laboratory at only 400°C.

The few palaeomagnetic data available for the Upper Cambrian Dundas Group of Tasmania (Briden, 1967*b*; Embleton & Giddings, 1974) are free of the inherent problems which complicate the South Australian data. The pole position is based on results from sedimentary and igneous rock types. The stability of the remanent magnetism was demonstrated by thermal and A.F. cleaning techniques, the presence of reversed polarities and the reduced scatter of directions after applying the fold test (Graham, 1949; McElhinny, 1964). The pole at 23°S, 13°E ($A_{95} = 11.5°$) agrees with poles of equivalent age from the rest of Australia. Early Ordovician fault drag or some other mechanism was therefore not

responsible for bending the Dundas Trough and the east-west portion of the Mount Reed volcanic arc from their supposed primary meridional trends.

CONCLUSIONS

We have concerned ourselves principally with the structural implications of large-scale horizontal displacements in southern Australia. The palaeomagnetic data presented bring into dispute that the effects of shearing produced rotations on the scale envisaged. We do not, however, dispute the geological evidence for activity along a structural lineament as described in the hypothesis. A relatively simple linear displacement of a few hundred kilometres would probably not be resolved palaeomagnetically.

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